Automated Driving Systems

Share:

NHTSA's mission is to save lives, prevent injuries, and reduce the economic costs of roadway crashes through education, research, safety standards and enforcement activity. Advanced vehicle technologies hold the promise not only to change the way we drive but to save lives.

The continuing evolution of automotive technology aims to deliver even greater safety benefits than earlier technologies. One day, automated driving systems, which some refer to as automated vehicles, may be able to handle the whole task of driving when we don't want to or can't do it ourselves.

NHTSA demonstrates its dedication to saving lives on our nation's roads and highways through its approach to the safe development, testing, and deployment of new and advanced vehicle technologies that have enormous potential for improving safety and mobility for all Americans.

AV 4.0

ENSURING AMERICAN LEADERSHIP IN AUTOMATED VEHICLE TECHNOLOGIES: AUTOMATED VEHICLES 4.0

LEARN MORE

ENSURING AMERICAN LEADERSHIP IN AUTOMATED VEHICLE TECHNOLOGIES AV 4.0 (PDF, 2.7...

AV 4.0, announced at CES in January 2020, builds upon *Preparing for the Future of Transportation: Automated Vehicles 3.0 (AV 3.0)* by expanding the scope to 38 relevant United States Government (USG) components which have direct or tangential equities in safe development and integration of AV technologies. AV 4.0 is structured around three key areas: USG AV principles, administration efforts supporting AV technology growth and leadership, and USG activities and opportunities for collaboration.

AV 4.0 seeks to ensure a consistent USG approach to AV technologies, and to detail the authorities, research, and investments being made across the USG so that the United States can continue to lead AV technology research, development, and integration.

AV 3.0

PREPARING FOR THE FUTURE OF TRANSPORTATION: AUTOMATED VEHICLES 3.0

GUIDANCE

PREPARING FOR THE FUTURE OF TRANSPORTATION: AUTOMATED VEHICLES 3.0

On October 4, 2018, the Department of Transportation released new Federal guidance for automated vehicles – *Automated Vehicles 3.0: Preparing for the Future of Transportation 3.0.* This builds upon DOT's *2.0: A Vision for Safety* and provides guidance for states to consider for the training and licensing of test drivers. It also offers guidance for testing entities to consider driver engagement methods during testing.

Additionally, NHTSA announced the ADS pilot, a preliminary step that seeks public comment on a national pilot research program to help safely test and deploy ADS-equipped vehicles. This collaboration could help aid the research and development of safety standards for advanced vehicle safety technologies.

- **Nov. 16, 2018**: Federal Register notice for extension of pilot ANPRM comment period

AV 3.0 offers our nation a plan to guide the development of exciting new technologies. As we embrace technological advances in motor vehicle transportation, safety must remain the top priority. Together, we can encourage new technologies that improve safety without hampering innovation.

NHTSA reports across a number of topics addressing new vehicle technologies. See and download some recent NHTSA reports a *Published Reports and Documents*.

- Read the news release announcing AV 3.0
- Download *Automated Vehicles 3.0: Preparing for the Future of Transportation 3.0*
- DOT's Automated Vehicles activities

AUTOMATED DRIVING SYSTEMS 2.0: A VISION FOR SAFETY

GUIDANCE

AUTOMATED DRIVING SYSTEMS 2.0 VOLUNTARY GUIDANCE

U.S. DOT and NHTSA analyzed Federal Register docket (NHTSA-2016-0090) comments, public meeting proceedings and other stakeholder discussions, recent Congressional hearings, and state activities, and used this analysis as the foundation for improvements and refinements to develop NHTSA's new voluntary guidance – *Automated Driving Systems 2.0: A Vision for Safety*. The new guidance is clearer, more streamlined, less burdensome and contains additional, more helpful information for states.

The length of this guidance is significantly reduced from 2016's *Federal Automated Vehicles Policy* and focuses on two sections: *Section I: Voluntary Guidance for Automated Driving Systems*; and *Section II: Technical Assistance to States, Best Practices for Legislatures Regarding Automated Driving Systems*.

The following provides a brief comparison of the new Voluntary Guidance and the FAVP.

Section I: Vehicle Performance Guidance for Automated Vehicles

2016 Federal Automated Vehicles Policy

Automated Driving Systems 2.0: A Vision for Safety

Model State Policy

2016 Federal Automated Vehicles Policy

Automated Driving Systems 2.0: A Vision for Safety

Voluntary Safety Self-Assessment Disclosure Index

Find the Voluntary Safety Self-Assessment (VSSA) template here, as well as a continuously updated index of VSSAs.

VSSA INDEX